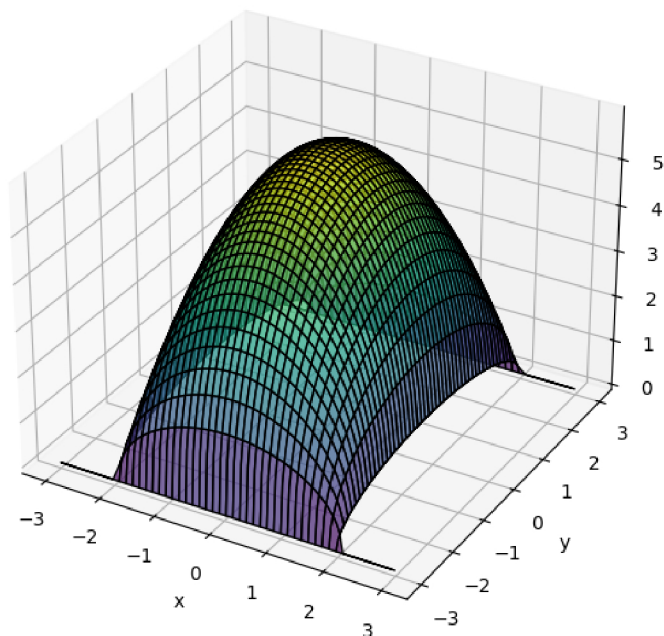




```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 x = np.linspace(-3, 3, 100)
5 y = np.linspace(-3, 3, 100)
6 X, Y = np.meshgrid(x, y)
7
8 Z = np.sqrt(((x**2)-4)*((Y**2)-9))
9
10 fig = plt.figure(figsize=(12, 6))
11
12 ax1 = fig.add_subplot(121, projection='3d')
13 ax1.plot_surface(X, Y, Z, cmap='viridis', edgecolor='k', alpha=0.7)
14 ax1.set_title("Surface Plot of  $f(x, y)$ ")
15 ax1.set_xlabel("x")
16 ax1.set_ylabel("y")
17 ax1.set_zlabel("z")
18
19 ax2 = fig.add_subplot(122)
20 contour = ax2.contour(X, Y, Z, levels=10, cmap='viridis')
21 ax2.clabel(contour, inline=False, fontsize=8)
22 ax2.set_title("Level Curves of  $f(x, y)$ ")
23 ax2.set_xlabel("x")
24 ax2.set_ylabel("y")
25
26 plt.tight_layout()
27 plt.show()
```

Surface Plot of $f(x, y)$



Level Curves of $f(x, y)$

